

TIMELY-Bench

Anchor-bounded temporal clinical reasoning in critical care

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Paper



Code



Contact



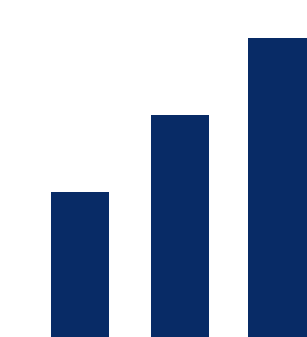
MIMIC-IV v3.1 and
MIMIC-IV-Note
66,485 task-eligible ICU stays
Frozen V3 benchmark release
Deduplicated across overlapping cohorts



4 condition families · 14 clinical tasks
AKI | Delirium | Sepsis | Stroke



168-hour ICU window
Evidence available through Anchor T
Post-anchor information withheld



12,000 sampled
benchmark instances
9,587 unique ICU stays



expanded over
eligible task-dimension
combinations



53,070 prompts per provider
full-multimodal · 9 frozen LLM providers

For timestamped prediction tasks, can models reason correctly using past observations?

Anchor-bounded prompts · synchronised A/B1/B2/B3 views · six-dimensional reasoning evaluation

1 Current Limitations



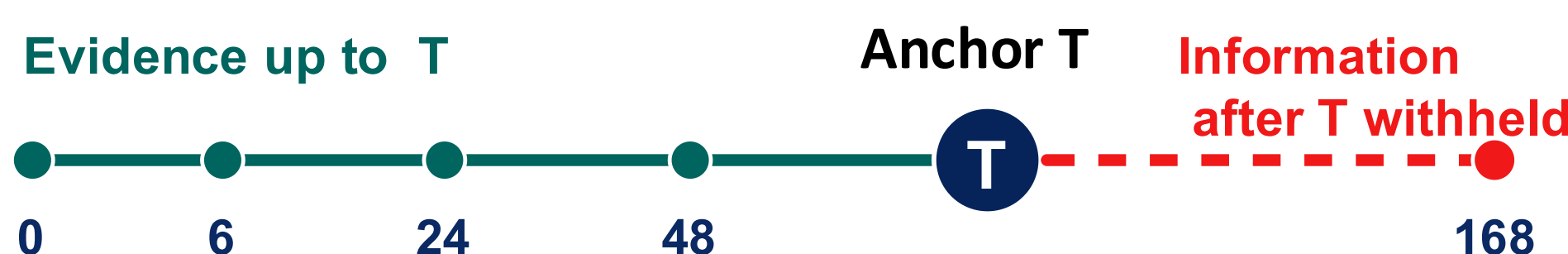
Static EHR labels can hide timing errors.



Long EHR context can leak future evidence.



Clinical reasoning requires timing, thresholds, trends, mechanisms, and evidence.



2 Research questions

- RQ1** Does preserving an explicit hourly sequence improve over a compact anchor-state summary?
- RQ2** How does reasoning difficulty vary by condition, reasoning dimension, and trajectory regime?
- RQ3** Does medical-domain adaptation improve anchor-bounded longitudinal EHR reasoning?

3 Data and tasks

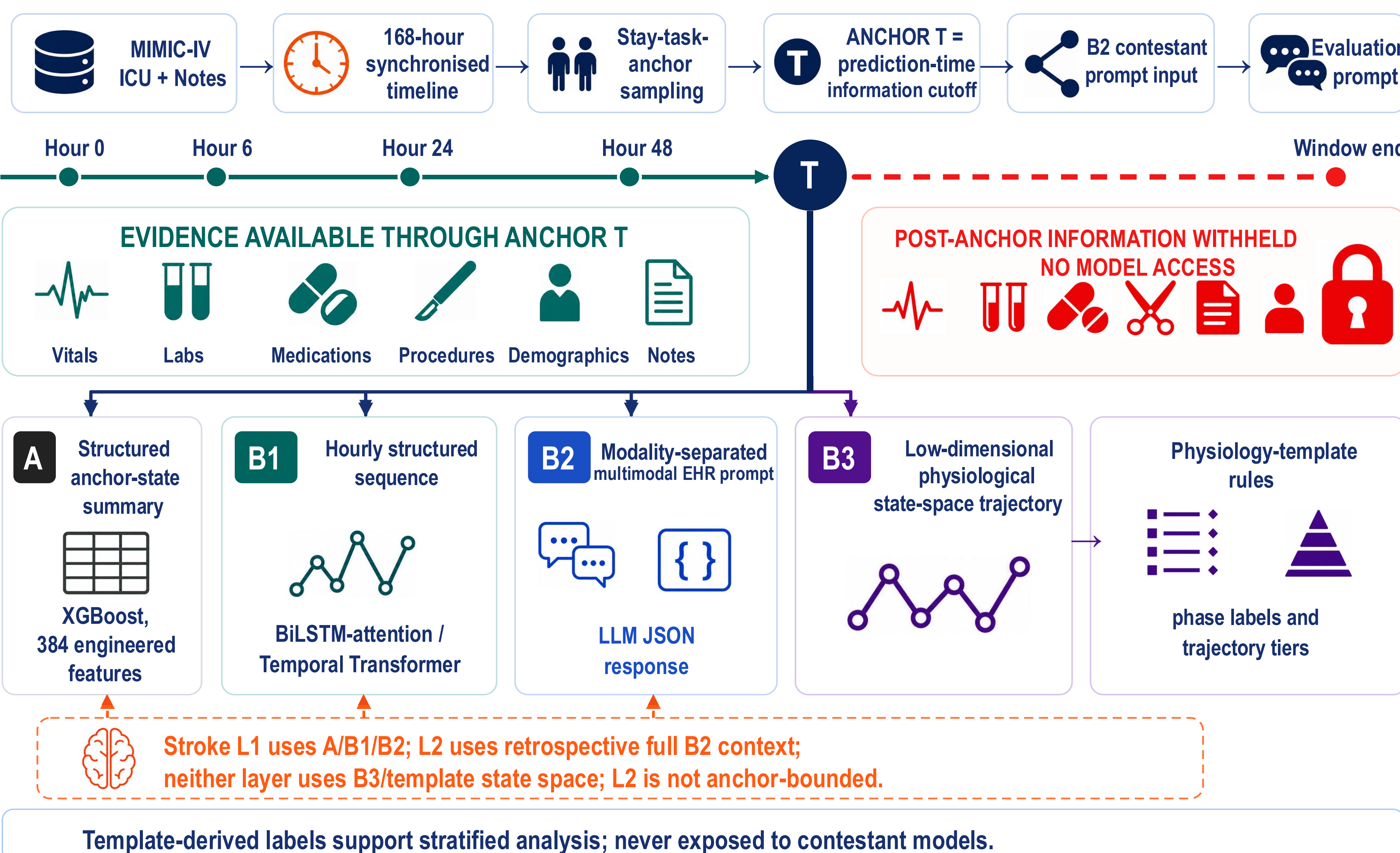
Condition	Stays*	Task IDs	Target and horizon	Key evidence
AKI	49,920	AKI-T1 / AKI-S1	Stage 2+ within 24 h; RRT proxy within 72 h	creatinine · urine output · RRT
Delirium	21,628	DEL-T1 / DEL-S1	persistence or resolution within 24 h	CAM-ICU · RASS · GCS
Sepsis	29,713	SEP-T1 / SEP-S1	First post-onset shock within 12 h; ≥10% lactate reduction within 6 h	MAP · vasopressors, lactate
Stroke	5,318	S-T1..T4 / S-R1..R4	L1 prediction; L2 retrospective synthesis / extraction	strength · laterality · imaging

*Condition-specific task cohorts overlap; deduplicated union = 66,485 stays.

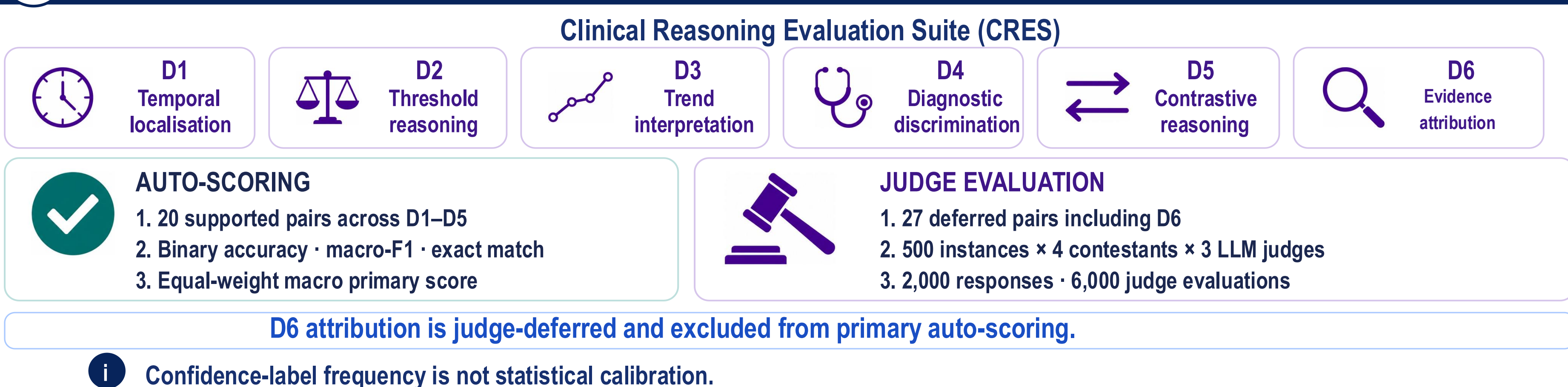
T = temporal | S = state/secondary trajectory | R = retrospective synthesis/extraction

Stratified sampling: maximum two anchors per stay per task where applicable.

4 Benchmark pipeline

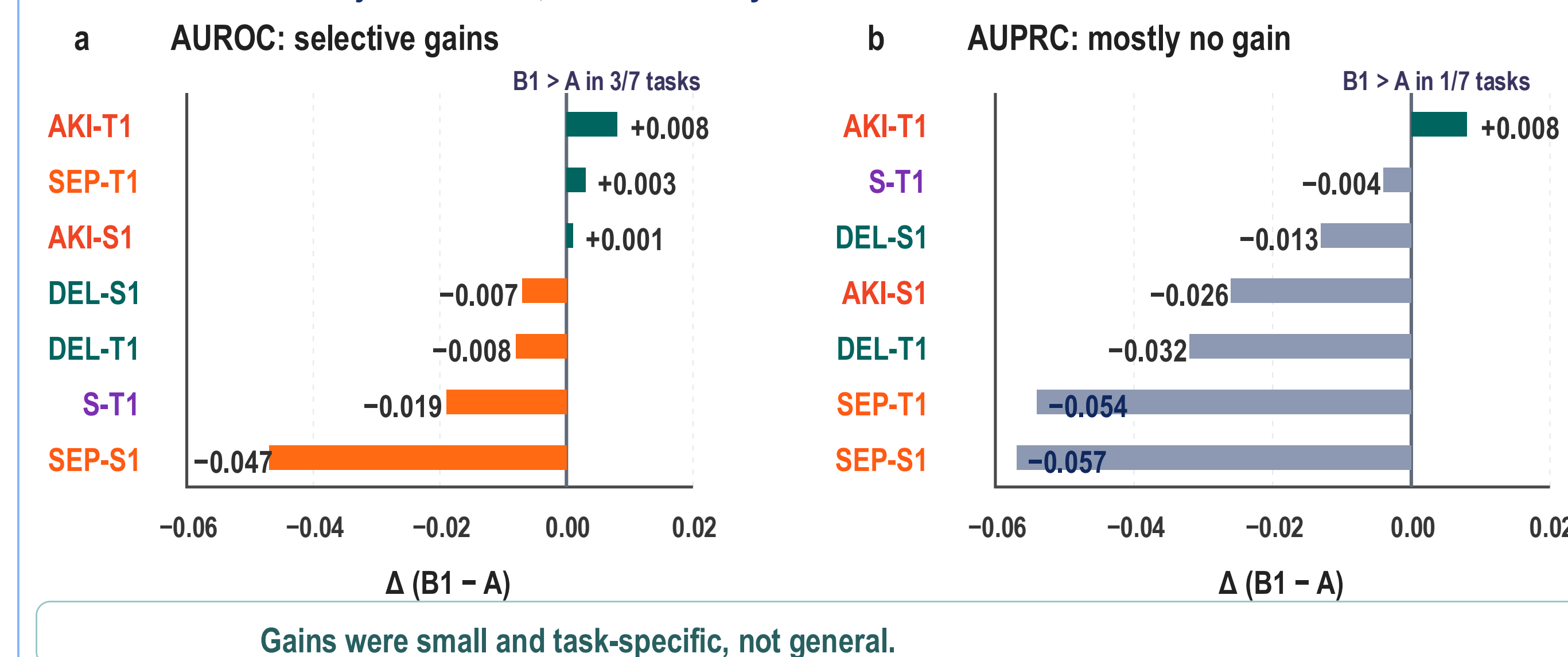


5 Evaluation

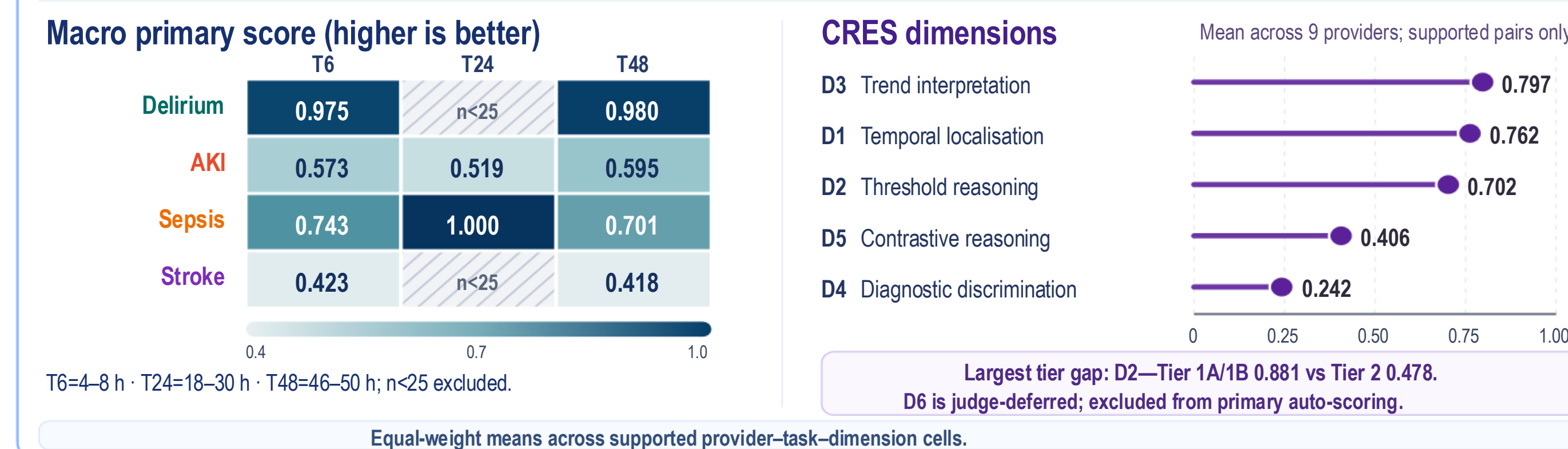


6 Results

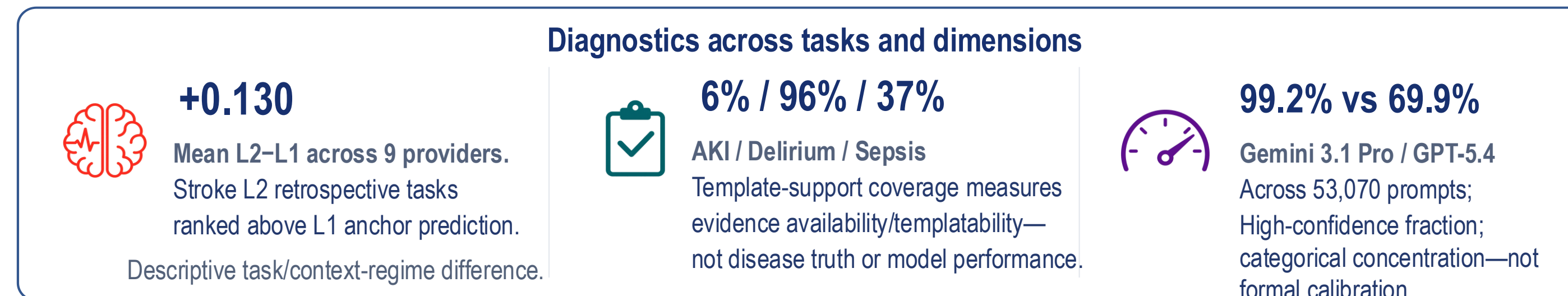
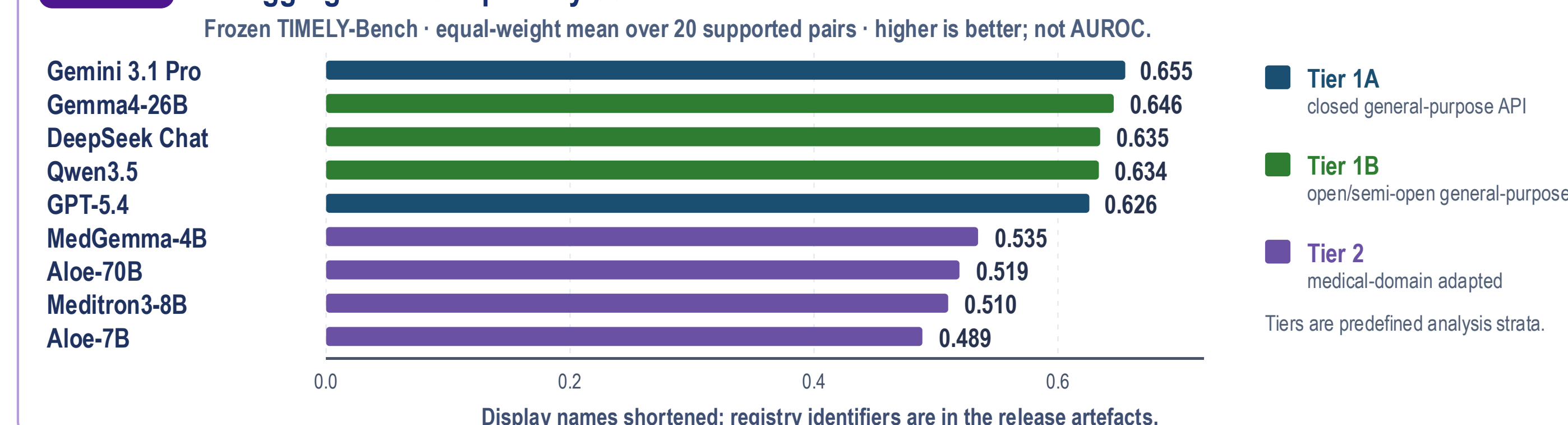
RQ1 Small and selective gains from hourly sequence modelling
B1 improved AUROC in 3/7 eligible binary tasks; AUPRC in 1/7.
5-fold subject-level CV; B1 selected by within-task AUROC.



RQ2 Difficulty depends on condition and reasoning dimension
Temporal sensitivity is non-monotonic; CRES failures are dimension-specific.



RQ3 General-purpose models ranked above the evaluated medical-domain models on aggregate macro primary score.



7 Conclusion

Temporal reasoning was condition- and operation-specific: sequence modelling helped selectively, while general-purpose LLMs led the aggregate ranking on TIMELY-Bench.



Anchor-bounded evaluation fixes the prediction-time information state.



Across supported pairs, trend interpretation scored higher than diagnostic discrimination.



Template support and stroke layers expose case-mix and task-regime differences.



RETROSPECTIVE, SINGLE-SOURCE LIMITATION

This MIMIC-IV v3.1 and MIMIC-IV-Note evaluation does not establish cross-institutional transportability, prospective workflow performance, or readiness for clinical deployment.